

SUMMARY

Mechanical Engineer with over two years of experience in design, simulation, and laboratory testing of electromechanical assemblies, medical devices, and manufacturing systems. Current responsibilities include developing haptic actuation systems and control logic, experimental design, data acquisition, visualization, and mechanical analysis. Additional experience in a marine engineering internship, computational fluid dynamics (CFD) projects, and technical writing (journal papers, thesis, grant proposals, and documentation). Former Division I baseball athlete and recipient of multiple engineering scholarships.

TECHNICAL SKILLS AND PROFICIENCIES

Engineering Skills: Actuation systems and control- *electric motors and controllers, PIDs*; Microcontrollers- *Arduino, ESP32, Raspberry Pi*; Sensors; Experimental design; Data acquisition and visualization- *Python, MATLAB, LABVIEW, DEWESoft*; Rapid prototyping- *3D printing, FDM, SLA, PCB design, protoboards*; Welding; Computer-aided design (CAD)- *SolidWorks, Rhino, Blender, Fusion 360*; Programming- *Python, C/C++, MATLAB, ROS2*; Computational fluid dynamics (CFD)- *Fluent, OpenFOAM, Orca3D*; Finite element analysis (FEA)- *ANSYS, ABAQUS, FEBio*;
Soft skills: Technical writing/documentation; Public speaking; Mentorship; Time management; Organization;

EXPERIENCE

Mechanical Engineering Research Assistant – CHROME Lab (St. Louis) Aug 2023 - Present
Wearable Haptic Integrated Mindfulness System (WHIMS)

- Performed requirement-based design cycles of nine devices in three user studies evaluating discriminative and affective haptic attributes
- Fabricated and bench tested electromechanical haptic actuation systems, achieving >90% localization rates
- Improved structure vibration dampening by 71% through FEA modal analysis and experimental validation using laser Doppler vibrometry

Additively Manufactured Blind Ice Hockey Enclosure

- Conducted mechanical durability and sound testing of four puck enclosures to optimize design performance
- Prototyped electromechanical noise emitters leveraging actuators, accelerometers, and wireless communication

Manufacturing Engineering Fellowship – Lighthouse for the Blind (St. Louis) June 2024 – Present

- Designing and adapting six accessible manufacturing technologies for blind and visually impaired employees
- Developing an auxiliary audio system for digital scales to output precise, non-visual measurements
- Adapting the native inkjet coder interface to be reprojected on an external monitor with color correction

Mechanical Engineering Intern – Regal Boats (Orlando, FL) May 2023 – Aug 2023

- Performed CFD analysis and validation on hull designs for running characteristics and excitation response
 - Data acquisition of boat vibration and running characteristics with DEWESOFT sensor and DAQ systems
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PROJECTS

External Ventricular Drain (EVD) Clearance – Washington University in St. Louis Dec 2024 – May 2026

- Designed and developed a novel vibration-based catheter clearance system, doubling drain obstruction time
- Characterizing 25 vibration propagation profiles and quantifying their effects on fluid clearance efficiency

Computational Fluid Dynamics (CFD) for Microfluidic Flow Device Jan 2024 – May 2024

- Leveraged simulation results for shear stress and fluid dynamics to guide device design and validate fluid conditions, ensuring physiologically relevant cell culture experiments

Infrared Photogate for Sickle Cell Confirmation- 2nd place Rice 360 Global Health Challenge Oct 2021 – May 2023

- Engineered a low cost (<\$25) automated data collection system for diagnosing Sickle Cell in developing regions
 - Conducted a multiphase CFD study in OpenFOAM to evaluate alignment impact on droplet dynamics
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EDUCATION

Master of Science (MS): Aerospace and Mechanical Engineering, Saint Louis University May 2026

- 4.00/4.00 GPA; Thesis- *Design and Evaluation of Wearable Haptic Devices for Non-Visual Guided Breathing*

Bachelor of Science (BS): Mechanical Engineering, Saint Louis University May 2024

- 3.99/4.00 GPA; *summa cum laude*; Dean's List 2020-2024

Scholarships and Awards

Keynote speaker at 2024 SLU scholarship dinner, Patrick P. Lee Engineering Scholarship, Gerald E. Dreifke Engineering Scholarship, Roland Quest Scholarship, Dr. Bellur L. Nagabhushan Scholarship, Division 1 Baseball Scholarship